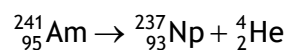


6. (a) The following statement represents a nuclear decay in which an alpha particle is emitted.



- (i) Explain why energy is released in this decay.

1

- (ii) The mass of the particles involved in this decay are shown in the table.

Particle	Mass (kg)
${}_{95}^{241}\text{Am}$	4.0028×10^{-25}
${}_{93}^{237}\text{Np}$	3.9363×10^{-25}
${}_2^4\text{He}$	6.6447×10^{-27}

Determine the energy released in this decay.

4

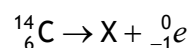
Space for working and answer



6. (continued)

- (b) Carbon-14 undergoes nuclear decay by releasing a beta particle ${}_{-1}^0e$.

The following statement represents this decay.



- (i) Identify the element represented by X.

1

- (ii) An explanation for beta decay is that a neutron decays into a proton and an electron. The electron is emitted as a beta particle.

However, studies of beta decay show that the mass-energy equivalence of carbon-14 is greater than the total mass-energy equivalence of element X and the beta particle.

State the conclusion that particle physicists have drawn from this evidence about what happens during beta decay.

1

[Turn over



6. (continued)

- (c) Quarks are fundamental particles that do not exist by themselves but make up other particles called hadrons.

There are six types of quark: up, down, charm, strange, top, and bottom.

- (i) Explain what is meant by the term *fundamental particle*.

1

- (ii) Hadrons can be further classified according to the number of quarks they contain.

State the type of particles that contain three quarks.

1



6. (c) (continued)

(iii) Scientists have discovered a type of particle called pentaquarks.

Pentaquarks contain 4 quarks and 1 antiquark.

Recently scientists at CERN reported the discovery of a pentaquark containing 1 up, 1 down, 1 strange, 1 charm, and 1 anticharm quark.

The table contains information about the charge of some of the quarks that make up this pentaquark.

Type of quark	Charge
Up	$+\frac{2}{3}e$
Down	$-\frac{1}{3}e$
Charm	$+\frac{2}{3}e$

The overall charge on the pentaquark is 0.

Determine the charge of a strange quark.

Space for working and answer

2

[Turn over

